

Data modelling

- It defines that is there any relationship between tables or not.
- Suppose If I load any group of tables, so if I want to take any single column from first table and another column from second table, then I have to build relation between 2 tables.

But now when I take visual and I take column from these tables so it is completely correct.

Reason:- My PBI have already build relationship between two tables.

But for study purpose will delete relationship between them.

Model view → Manage relationship → Check the box from table
→ Delete → Close

Now, Can we create relationship by dragging column one by one but if I want to make it automatic then again go to manage relationship.

Manage relationship → Auto detect → Close

Types of tables:-

- (I) Dimension table
- (II) Fact table

* Primary keys:- It gives the unique identification to each row and it consist of unique values.

* Foreign key:- It helps to create relationship with other table and it may consist repeated and null value.

(1) Fact table $\Rightarrow$ It consist the foreign key and numeric columns, like sales, profit etc.

(II) Dimension table $\Rightarrow$ It consist the primary key it explain fact table.

* Fact table $\rightarrow$ Foreign Key

* Dimension table $\rightarrow$ Primary Key

Example $\Rightarrow$ I am having, C-Id, C-Name, products in one table and C-ID, sales, profit, in another table.

So my first table is DT (dimension table) and next table FT (Fact table)

Schema

$\Rightarrow$ It is defined as. "How the tables are connected with each other" in data modelling.

$\Rightarrow$ It show any arrangement not relation. Relation is shown by Cardinality.

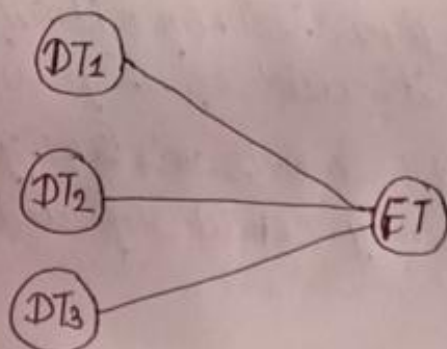
* Types:-

(I) Star Schema.

(II) Snowflake Schema.

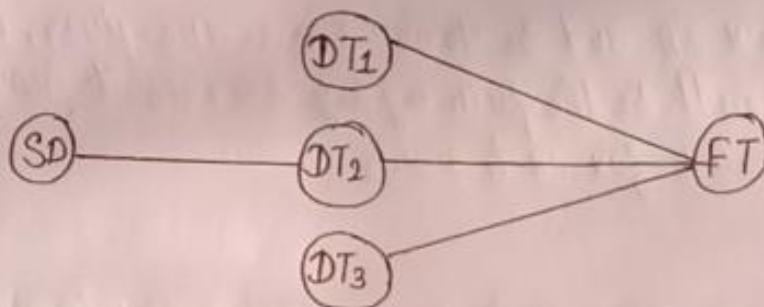
(III) Galaxy Schema (Fact constellation)

(I) Star Schema:



- ⇒ In star schema, all the dimension table is directly connected to fact table, it consist single fact table and multi dimension table.
- ⇒ It follows denormalisation / non redundancy / duplicate can consist.
- ⇒ No dimension table are interlinked.

(II) Snow-flake schema:

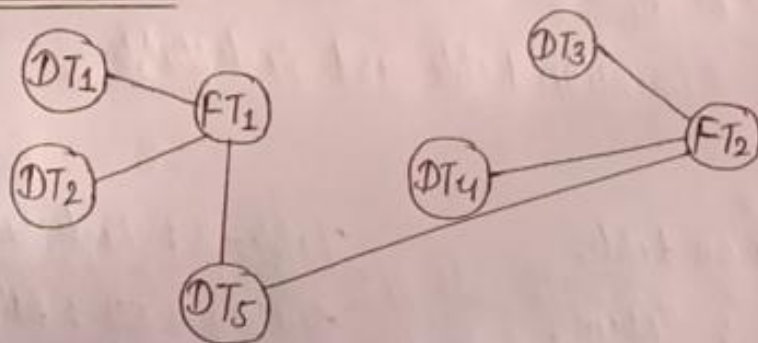


- ⇒ If the dimension table are interlink with each other or any dimension table is connected to its sub dimension table (SD)

* sub-dimension table:

It is sub part of dimension table.

(III) Galaxy schema:



- ⇒ It consist more than 1 fact table and at least one common dimension is in between fact table.